Answer **one** question from this section in the spaces provided.

8 A theme-park ride consists of two cages. They are moving in a circular path at constant speed v about a horizontal axis. Fig. 8.1 shows the ride at one instant when cage A is vertically above cage B.

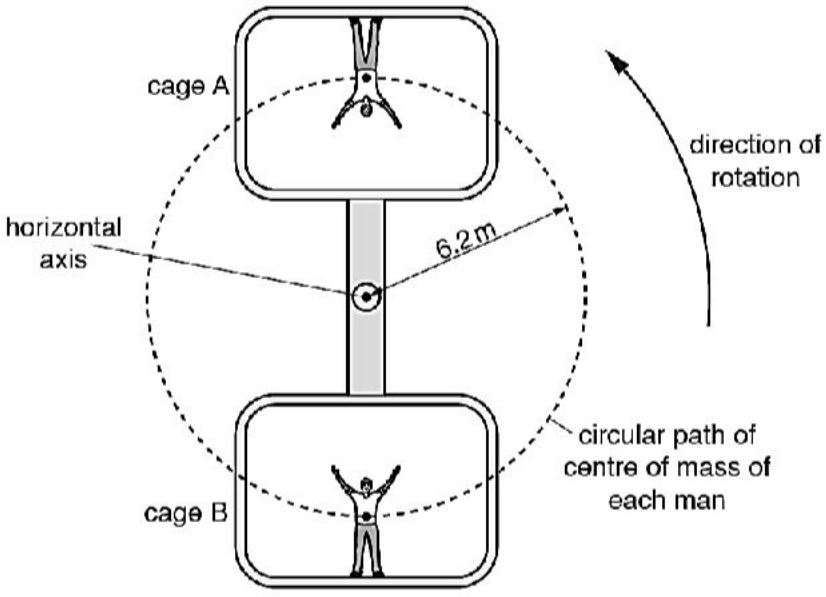


Fig. 8.1

(a) A man is riding in each cage. The mass of each man is 75 kg. The centre of mass of each man is 6.2 m from the horizontal axis. The period of one rotation is 4.1 s.

(i) Determine the speed v of the centre of mass of each man.

$$v = r\omega$$
$$= r\left(\frac{2\pi}{T}\right)$$
$$= 6.2\left(\frac{2\pi}{4.1}\right) \quad [1]$$
$$= 9.5 \text{ m s}^{-1} \quad [1]$$

speed v = ... m s^{-1} [2]

(ii) Calculate the magnitude of the acceleration of the centre of mass of each man.

$$a = r\omega^2$$
$$= r\left(\frac{2\pi}{T}\right)^2$$
$$= 6.2\left(\frac{2\pi}{4.1}\right)^2 \quad [1]$$
$$= 14.6 \text{ m s}^{-2} \quad [1]$$

acceleration = ... m s^{-2} [2]

(b) Fig. 8.2 shows the forces acting on the man in cage B at the instant the cages are in the positions shown. It shows the man in cage A at that same instant.

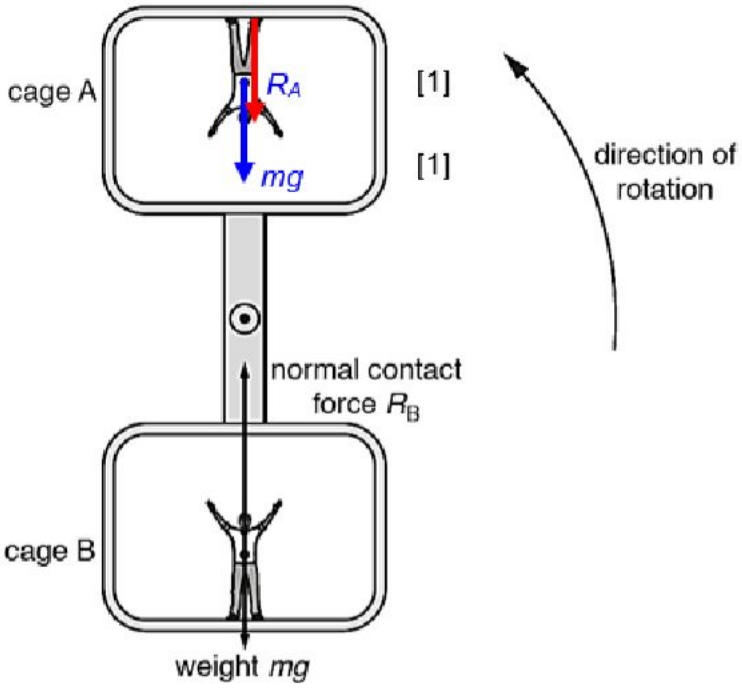


Fig. 8.2

(i) On Fig. 8.2, draw labelled arrows to represent the magnitude and direction of the forces acting on the man in cage A. [2]

(ii) Calculate the magnitude of the normal contact force R_A on the man in cage A at this instant.

$mg + R_A = ma_c$ [1]

$R_A = ma_c - mg$

$= 75(14.6 - 9.81)$

$= 360$ N (2 sf) [1]

R_A = ... N [2]

(c) **(i)** Explain why a minimum value for the speed is needed for the man in cage A to maintain in contact with the floor of his cage.

$R = ma - mg$, $R > 0$ is required to remain contact. [1]

so $a > g$, i.e. $v^2 / r > g$

$v > \sqrt{rg}$, the minimum speed for a contact force to be required [1] [2]

(ii) Calculate this minimum speed.

From (i), $v_{min} = \sqrt{rg}$

$= \sqrt{6.2 \times 9.81}$

$= 7.8$ m s^{-1} [1]

minimum speed = ... m s^{-1} [1]

(d) (i) State Newton's Law of Gravitation.

NLG states that two point masses attract each other with a force directly proportional to the product of the masses [1], and inversely proportional to the square of the distance between them. [1] [2]

(ii) The astronomer Johannes Kepler showed that the period T of rotation of a planet about the Sun is related to its mean distance R from the centre of the Sun by the expression

$$\frac{R^3}{T^2} = k$$

where k is a constant.

Show that, for planets in circular orbits about the Sun of mass M, the constant k is given by

$$k = \frac{GM}{4\pi^2}$$

where G is the gravitation constant.

Gravitational force provides the centripetal force [1]

$$\frac{GMm}{R^2} = mR\omega^2 \quad [1]$$

$$\frac{GMm}{R^2} = mR\left(\frac{2\pi}{T}\right)^2 \quad [1]$$

$$\frac{R^3}{T^2} = \frac{GM}{4\pi^2}$$

[3]

(iii) Earth can be assumed to be in a circular orbit about the Sun with an orbital radius of 1.50×10^8 km.

Use the expression in **d(ii)** to calculate a value for the mass of the Sun.

$$\frac{R^3}{T^2} = \frac{GM}{4\pi^2}$$

$$\frac{(1.50\times10^{11})^3}{(365\times24\times60\times60)^2} = \frac{6.67\times10^{-11}\times M}{4\pi^2} \quad [1]$$

$$M = 2.01\times10^{30} \text{ kg} \quad [1]$$

mass of Sun = ... kg [2]

(iv) Explain why, although planets experience gravitational forces from the Sun, they do not fall into the Sun.

At any instant, each planet is moving in a direction tangential to its orbit.[1] According to Newton's first law, the planet wants to continue to move in this direction [1] instead of falling into the Sun.

[2]

[Total: 20]